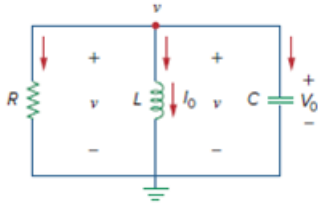


Problem

In Fig. 8.13, let $R = 2\ \Omega$, $L = 0.4\ \text{H}$, $C = 25\ \text{mF}$, $v(0) = 0$, $i(0) = 50\ \text{mA}$. Find $v(t)$ for $t > 0$.

Figure 8.13: A source-free parallel RLC circuit.



Step-by-step solution

Step 1 of 6

Draw the circuit diagram:

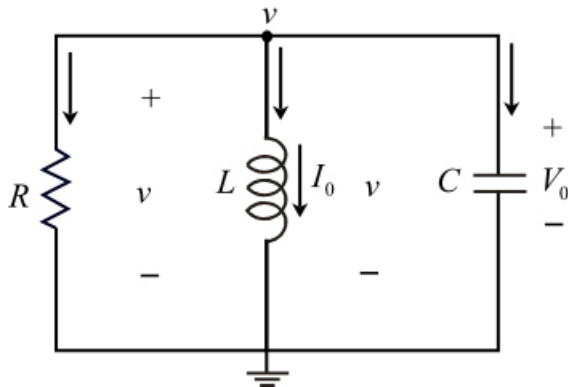


Figure 1

Step 2 of 6

Apply Kirchhoff's current law at the top node in Figure 1.

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(t) dt + C \frac{dv}{dt} = 0$$

Differentiate the equation with respect to ' t '.

$$\frac{1}{R} \frac{dv}{dt} + \frac{1}{L} \frac{d}{dt} \left(\int_{-\infty}^t v(t) dt \right) + C \frac{d^2v}{dt^2} = 0$$

$$\frac{d^2v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

Step 3 of 6

Write the characteristic equation.

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$$

Calculate the neper frequency.

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{(2)(2)(25 \times 10^{-3})} \\ &= 10\end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(0.4)(25 \times 10^{-3})}} \\ &= 10\end{aligned}$$

Step 4 of 6

Calculate the roots of a characteristic equation.

$$\begin{aligned}s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -10 \pm \sqrt{10^2 - 10^2} \\ &= -10\end{aligned}$$

Since $\alpha = \omega$, we have the critically damped case.

Hence, the response is

$$\begin{aligned}v(t) &= (A_1 + A_2 t)e^{-\alpha t} \text{ V} \\ v(t) &= (A_1 + A_2 t)e^{-10t} \text{ V} \dots\dots (1)\end{aligned}$$

Substitute 0 for t in equation (3).

$$\begin{aligned}v(0) &= (A_1 + A_2(0))e^{-10 \times 0} \text{ V} \\ 0 &= A_1 \\ A_1 &= 0\end{aligned}$$

Step 5 of 6

Apply Kirchhoff's current law at top node of Figure 1.

$$\frac{v(t)}{R} + i(t) + C \frac{dv(t)}{dt} = 0$$

$$C \frac{dv(t)}{dt} = -\frac{v(t)}{R} - i(t)$$

$$\frac{dv(t)}{dt} = -\frac{v(t)}{RC} - \frac{i(t)}{C}$$

$$\frac{dv(t)}{dt} = -\left(\frac{v(t) + Ri(t)}{RC}\right)$$

Substitute 0 for t in equation.

$$\begin{aligned}\frac{dv(0)}{dt} &= -\left(\frac{v(0) + Ri(0)}{RC}\right) \\ &= -\left(\frac{0 + (2)(50 \times 10^{-3})}{(2)(25 \times 10^{-3})}\right) \\ &= \frac{-100 \times 10^{-3}}{50 \times 10^{-3}} \\ &= -2 \text{ V/s}\end{aligned}$$

Step 6 of 6

Differentiate the equation (1) with respect to ' t '.

$$\frac{dv(t)}{dt} = (A_1 + A_2 t)e^{-10t}(-10) + e^{-10t}A_2$$

Substitute 0 for t in equation.

$$\begin{aligned}\frac{dv(0)}{dt} &= A_1(-10) + A_2 \\ -2 &= (-10)(0) + A_2 \\ A_2 &= -2\end{aligned}$$

Substitute 0 for A_1 and -2 for A_2 in equation (1)

$$\begin{aligned}v(t) &= (A_1 + A_2 t)e^{-at} \\ &= (0 - 2t)e^{-10t} \\ &= -2te^{-10t} \text{ V}\end{aligned}$$

Hence the voltage across the capacitor is $v(t) = -2te^{-10t} \text{ V}$ for $t > 0$.